

Multimodal RAG for Leishmaniasis Diagnosis

Traceable evidence, model-dependent retrieval, and safety-gated demonstration

Nguyen Tuan Ngoc · Supervisor: Prof. Dr. Andreas Müller · Co-Supervisor: Dr. Tran Duc Khanh · May 2026

Leishmaniasis is not simple image classification

The Problem

- Overlapping CL, MCL, VL, PKDL, and non-leishmaniasis mimics; lab confirmation is often the decisive evidence.

Evidence Streams

Text

Symptoms

Image

Context

Lab

Weak reference

Why It Is Hard

- Case reports mix symptoms, captions, pathology, treatment, and author conclusions. Image appearance alone is insufficient.

SUPPORT IMAGE

PMC3140149 · PKDL/HIV case report



Infiltrated plaques on nose, wrist, and finger — useful visual context, not standalone ground truth.

Source: open-access PMC case-report image

Four questions, one thread: does evidence help safely?

Each RQ asks whether more evidence improves the pipeline without creating unsupported claims.

RQ1

Weak-Reference Evaluation

Can structured case reconstruction support scored diagnosis evaluation on held-out leishmaniasis cases?

RQ2

Small-Model Practicality

Can local open models run on a single GPU with enough latency and memory transparency for audit?

RQ3

Multimodal Evidence

Do images and image-linked context add useful signal, or mainly increase visual hallucination risk?

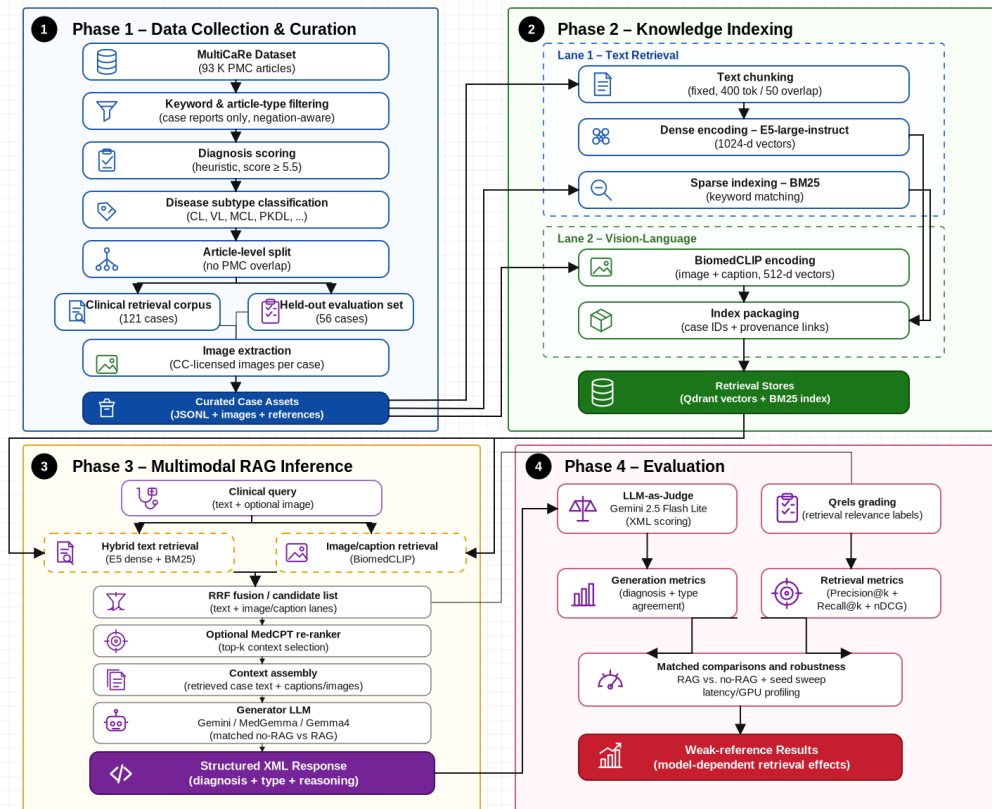
RQ4

Retrieval-Augmented Reasoning

Does retrieving similar cases help generation compared with matched no-RAG conditions?

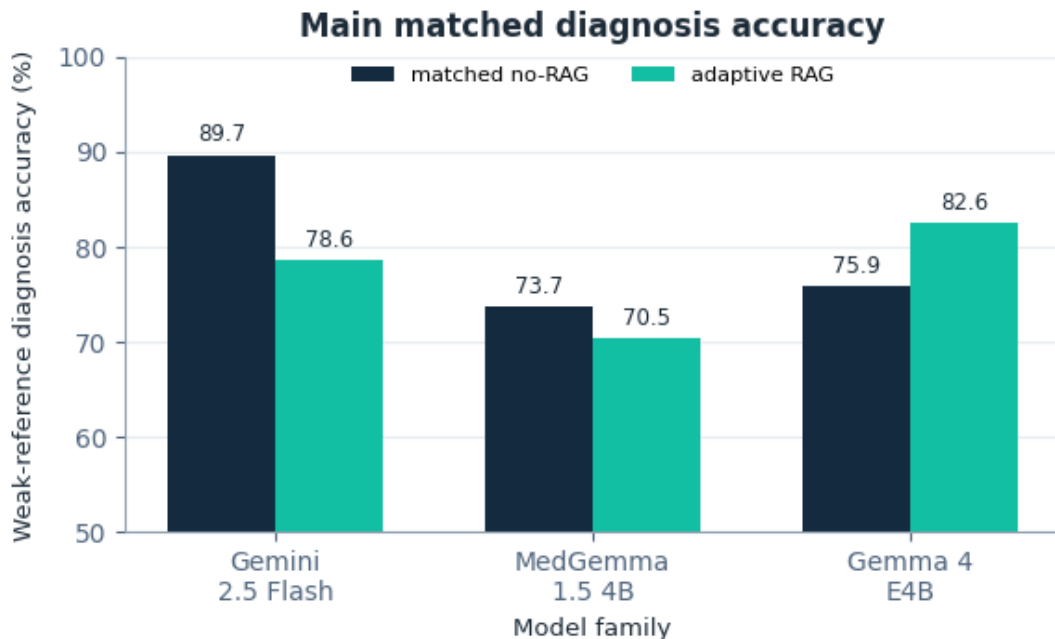
Two-lane retrieval architecture

Text retrieval and image-linked retrieval are fused, gated, and audited before presentation.



Main result: RAG effect is model-dependent

Weak-reference, prompt-judged diagnosis accuracy on 56 held-out cases; x-axis = model family, y-axis = diagnosis accuracy (%).



Key deltas

Gemini **-11.2 pp**

statistically supported degradation

MedGemma **-3.1 pp**

directional; not robust

Gemma 4 **+6.7 pp**

directional; CI crosses 0

Gemma 4 + images **+1.8 pp**

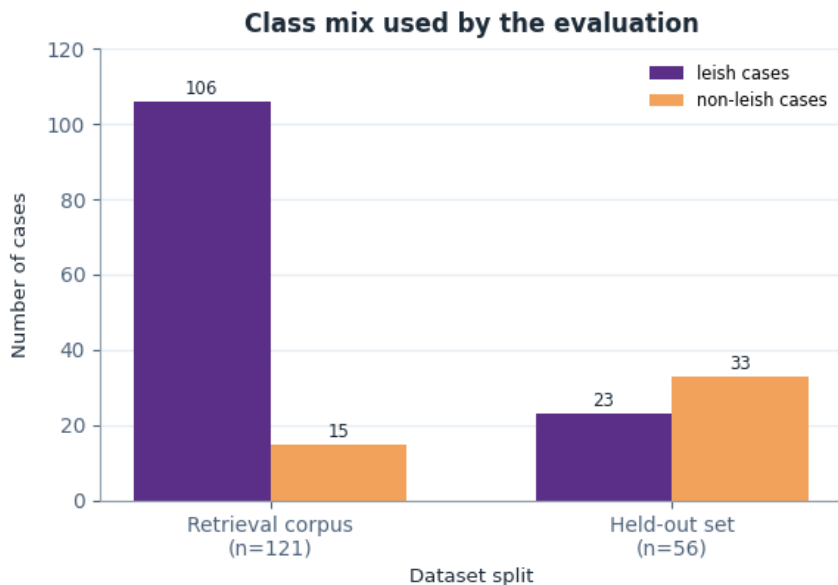
exploratory; hallucination risk ↑

Chart: grouped bars compare matched no-RAG vs adaptive RAG for each model family. Retrieval is not uniformly beneficial.

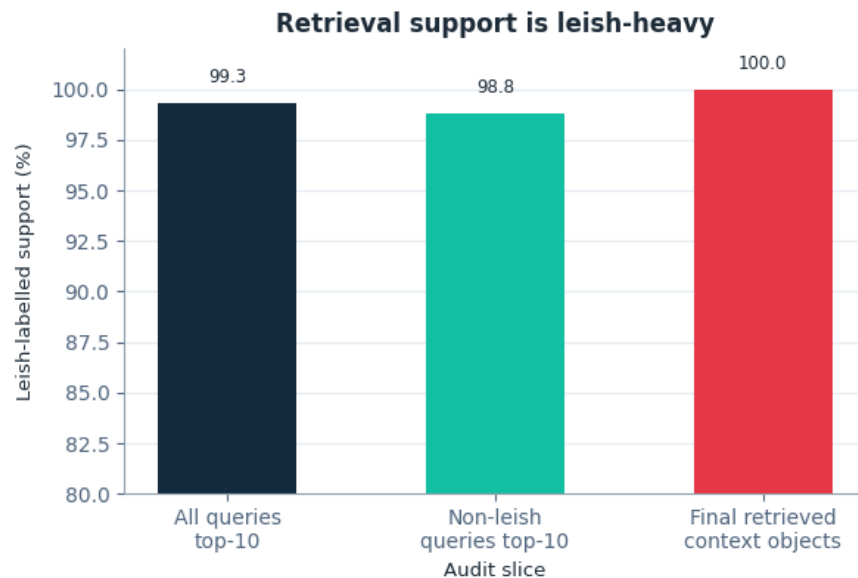
Corpus imbalance explains the specificity risk

Left chart: class mix by split. Right chart: leish-labelled share of retrieved support after clean-context audit.

Retrieval corpus



Held-out / support audit



Key finding: the corpus is 106/121 leish cases while the held-out set is majority non-leish. Non-leish queries still receive mostly leish-labelled top-10 support, so RAG can over-support leishmaniasis mimics.

Retriever quality sets the ceiling

Classic retrieval metrics on 23 held-out queries with qrels; 121-case corpus with 104-doc qrels coverage.

58.3%

P@5

Top-5 precision

59.5%

nDCG@5

Ranking quality @5

71.5%

MRR

First relevant hit

13.5%

Recall@5

Coverage in top 5

23.4%

Recall@10

Coverage in top 10

26.1%

MAP

Overall ranking

Top-rank precision and first-hit quality support explanations, but low recall means relevant cases often fall outside the prompt context window.

Image evidence: useful but risky

Chart axes now show model family (x-axis) and unsupported visual claims as a percentage (y-axis).

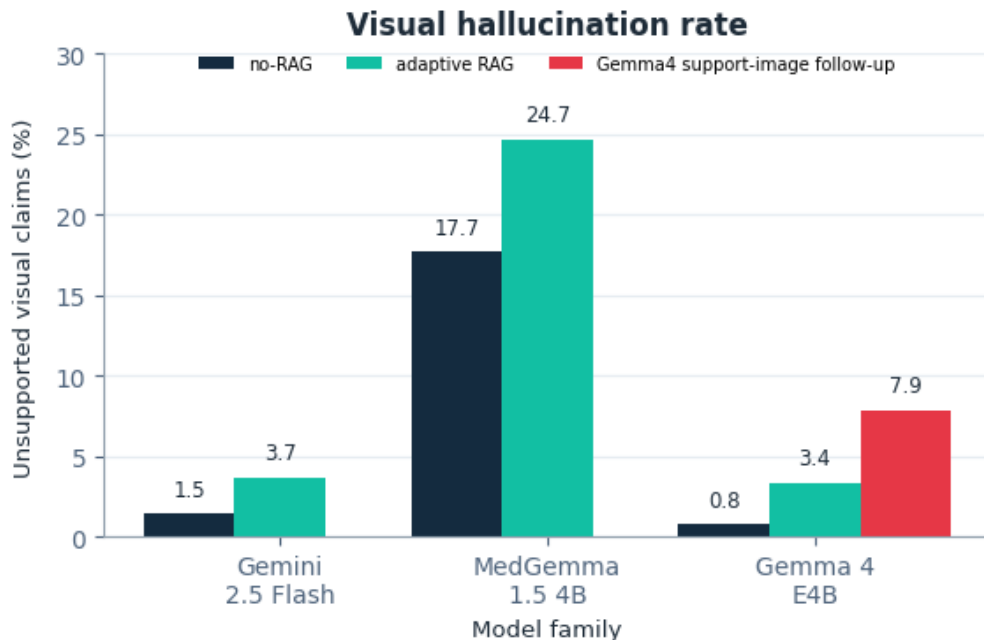


Image-evidence findings

✓ Images are context, not ground truth: captions, article text, and lab evidence still carry the diagnosis.

✗ PMC3140149 is a support/retrieval example, not a direct official test query.

✓ Gemma 4 support-image follow-up was directionally stronger than standard RAG, but the safety cost is visible in the red bar.

! Unsupported visual claims rose from 3.4% to 7.9% for the Gemma4 support-image follow-up.

✗ Adding images does not automatically improve diagnosis across models.

Visual hallucination = judge-detected unsupported visual/pathology claims. This is not expert adjudication, so the safe demo shows provenance and gate decisions.

Small-model practicality for a live defense

Single 24 GB GPU local models vs. Gemini API: latency, memory, auditability, and demo reliability.

Gemini 2.5 Flash Lite

API · external

Latency: 3–4 s/case
Memory: API/cloud
Best diag. acc.: 89.7%

Strengths

- ✓ Fastest inference
- ✓ Highest no-RAG accuracy
- ✓ Strong multimodal

Limitations

- ✗ RAG degraded under matched evaluation
- ✗ No local GPU audit trail
- ✗ External dependency

MedGemma 1.5 4B

Local · medical-tuned

Latency: 22–26 s/case
Memory: ~16 GB GPU
Best diag. acc.: 73.7%

Strengths

- ✓ Medical domain tuning
- ✓ Full auditability
- ✓ Privacy-safe

Limitations

- ✗ High visual-hallucination rate
- ✗ RAG directionally worsened
- ✗ Weaker reasoning in this setting

Gemma 4 E4B

Local · open

Latency: 46–59 s/case
Memory: ~18–19 GB GPU
Best diag. acc.: 82.6% (RAG)

Strengths

- ✓ Best directional RAG delta (+6.7 pp)
- ✓ Real GPU sidecar demo available
- ✓ Auditable

Limitations

- ✗ Slowest inference
- ✗ Directional, not conclusive
- ✗ Image evidence raises hallucination risk

Limitations are part of the contribution

The thesis claims are deliberately narrow: traceable research evaluation, not bedside-use evidence.

Weak-reference labels

No clinician adjudication; prompt-judged diagnosis accuracy is useful for controlled comparison, not clinical validation.

Corpus imbalance

The retrieval corpus is leishmaniasis-heavy, so non-leishmaniasis mimics remain a specificity stress test.

Small held-out set

Only Gemini's negative RAG delta is statistically supported; Gemma 4's +6.7 pp is directional only.

Adaptive RAG is route-mixed

Some rows received retrieved context and others fell back. Route-level audit is needed before interpreting aggregate accuracy.

Future: clinician-reviewed labels · balanced mimic set · stronger recall · stricter visual-claim gating · deployment-grade validation

Conclusion: feasible, traceable, not deployment-ready

RQ1 Feasibility

Research pipeline works

Structured cases, retrieval, generation, gate audit, and live demo are reproducible and traceable.

RQ2 Small Models

Practical but bounded

Local models run on a 24 GB GPU, but latency and memory make full live evaluation risky.

RQ3 Multimodal

Useful, risky signal

Image-linked evidence can help, but unsupported visual claims require conservative gating.

RQ4 RAG Effect

Model-dependent

Gemma 4 shows a positive directional RAG pattern; Gemini and MedGemma warn against universal RAG claims.

Final claim: multimodal RAG for leishmaniasis can be evaluated traceably under constrained resources; stronger claims require balanced data, stronger labels, and expert review.

What v6 was missing: core result map

v6 was compact for a live supervisor demo; thesis_v44b-1 contains a richer result chain that needs a backup map.

Main table set

5.1 main metrics; 5.2-5.3 clean-context; 5.5 modality; 5.6 retrieval; 5.7 route audit; 5.8 imbalance; 5.9 matched deltas.

Why v6 felt short

It compressed many result families into slides 5-10 to preserve a 30-minute defense flow.

What v3 adds

Labeled charts plus three backup slides for result provenance, clean-context sensitivity, routing, modality, latency, and GPU evidence.

Claim boundary

The backup slides explain existing thesis evidence; they do not create new experimental claims.

Use this slide when supervisors ask where each result family lives in thesis_v44b-1.

Clean-context and route audit explain the RAG effect

These are the missing mechanism checks behind the short v6 RAG story.

Clean-context deltas

Gemini -7.1 pp; MedGemma +6.7 pp; Gemma4 +16.1 pp. Same 56 case IDs, sanitized query artifact; supplementary sensitivity only.

Prompt2 A+B tier

47 cases: Gemini -4.3 pp; MedGemma +4.3 pp; Gemma4 +14.9 pp. Descriptive, not confirmatory.

Route-mixed RAG

Gemma4 audit: 31/56 adaptive_rag rows and 25/56 no-context fallback rows. RAG did not inject context into every case.

Retrieval imbalance

Corpus 106/121 leish; held-out 33/56 non-leish. Non-leish top-10 support was still 98.8% leish-labelled.

Takeaway: retrieval helps conditionally because generator behavior, route policy, and corpus balance all matter.

Sensitivity: modality, latency, and GPU practicality

This slide collects appendix evidence that supports the demo story without overclaiming deployment readiness.

Modality matrix

Gemini strongest multimodal no-RAG: 89.7%. Gemma4 text RAG: 85.3%; multimodal RAG: 82.6%; tensor-on RAG: 84.4% supplementary.

Image-risk trade-off

Gemma4 support images: +1.8 pp over standard RAG but visual hallucination rose 3.4% -> 7.9%. Multi-seed +8.3 pp vs no-RAG, CI crosses 0.

Local practicality

MedGemma: 22-26 s/case, peak 15.70 GB. Gemma4: 46-59 s/case, peak 18.84 GB. Gemini API: 3-4 s/case.

Defense point

The contribution is traceability, auditability, and bounded evidence, not a deployment-ready classifier.

These are the missing core result families behind the short v6 narrative.